

## Module 2 purpose: convert damage into electric field

Module 2 is the electrostatic bridge between radiation-induced defects and carrier transport.

$$\{C_i, z_i, n, p, N_D^+, N_A^-, \text{bias/BCs}\} \longrightarrow \rho(x, y, t) \longrightarrow \phi(x, y, t) \longrightarrow \mathbf{E}(x, y, t)$$

- ▶ The central unknown is the electrostatic potential  $\phi(x, y, t)$  in a 2-D silicon domain  $\Omega$ .
- ▶ Radiation defects enter through their concentrations  $C_i(x, y, t)$  and effective charge states  $z_i$ .
- ▶ The electric field is not solved independently; it is recovered after the potential solve:

$$\mathbf{E}(x, y, t) = -\nabla\phi(x, y, t).$$

- ▶ Module 2 supplies  $\phi$  and  $\mathbf{E}$  to carrier drift-diffusion, recombination, degradation metrics, and later coupled multiphysics updates.

**Physical interpretation:** charged defects perturb the local space charge; space charge bends the electrostatic potential; the potential gradient drives carrier motion.

# Governing electrostatics and defect space charge

## Electrostatic limit of Maxwell's equations

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho, & \mathbf{D} &= \epsilon \mathbf{E}, & \mathbf{E} &= -\nabla \phi, \\ \Rightarrow \quad \nabla \cdot (\epsilon \nabla \phi) &= -\rho. \end{aligned}$$

## Total semiconductor space charge

$$\rho(x, y, t) = q[p(x, y, t) - n(x, y, t) + N_D^+(x, y) - N_A^-(x, y)] + \rho_{\text{def}}(x, y, t),$$

$$\rho_{\text{def}}(x, y, t) = q \sum_{i=1}^M z_i C_i(x, y, t).$$

- ▶  $p$  and ionized donors  $N_D^+$  contribute positive charge;  $n$  and ionized acceptors  $N_A^-$  contribute negative charge.
- ▶ A defect species contributes positive, negative, or neutral charge according to the sign and magnitude of  $z_i$ .
- ▶ For spatially uniform silicon permittivity,

$$\epsilon = \epsilon_{\text{Si}}, \quad \epsilon_{\text{Si}} \nabla^2 \phi = -\rho.$$

- ▶ In nonuniform media or at material interfaces, the conservative form  $\nabla \cdot (\epsilon \nabla \phi) = -\rho$  must be retained.

# Finite-element weak form and Module 2 outputs

Weak form on the 2-D domain  $\Omega$

$$\int_{\Omega} \epsilon \nabla w \cdot \nabla \phi \, d\Omega = \int_{\Omega} w \rho \, d\Omega + \int_{\Gamma_N} w g_N \, d\Gamma$$

$$\phi_h(x, y, t) = \sum_{j=1}^N \Phi_j(t) N_j(x, y),$$

$$K_{ij} = \int_{\Omega} \epsilon \nabla N_i \cdot \nabla N_j \, d\Omega,$$

$$b_i(t) = \int_{\Omega} N_i \rho(x, y, t) \, d\Omega + \int_{\Gamma_N} N_i g_N \, d\Gamma,$$

$$K\Phi(t) = \mathbf{b}(t), \quad \mathbf{E}_h|_e(t) = -\nabla \phi_h|_e(t).$$

- ▶ Dirichlet boundaries impose contact voltages or a reference ground:  $\phi = \phi_D$  on  $\Gamma_D$ .
- ▶ Neumann boundaries impose normal electric displacement or insulating/symmetry conditions:  $-\epsilon \nabla \phi \cdot \hat{\mathbf{n}} = g_N$  on  $\Gamma_N$ .
- ▶ Main outputs: nodal potential  $\Phi$ , reconstructed field  $\mathbf{E}_h$ , and residual diagnostics.
- ▶ Module 2 is quasi-static: at each coupled time level, the current charge density determines the instantaneous electrostatic solution.

## Module 3 purpose: evolve the defect landscape

Module 3 converts the initial radiation damage map into a time-dependent defect concentration field.

Module 1 damage map  $\rightarrow C(x, y, 0)$  or  $S(x, y, t) \rightarrow C(x, y, t) \rightarrow$  Modules 2, 4, and 5

### Local defect balance law

$$\begin{aligned}\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J}_C &= R(C, T, \dots), \\ \mathbf{J}_C &= -D \nabla C, \\ \Rightarrow \quad \frac{\partial C}{\partial t} &= \nabla \cdot (D \nabla C) + R(C, T, \dots).\end{aligned}$$

- ▶  $C(x, y, t)$  is a coarse-grained concentration of radiation-induced defects in the 2-D silicon domain.
- ▶  $\mathbf{J}_C$  is the defect flux; Fick's law sends defects down the concentration gradient.
- ▶  $R$  contains local creation, removal, recombination, clustering, dissociation, or transformation.
- ▶ For multiple defect species,

$$\frac{\partial C_i}{\partial t} = \nabla \cdot (D_i \nabla C_i) + R_i(C_1, C_2, \dots, T).$$

# Diffusion, annealing, temperature dependence, and boundaries

## Baseline single-species diffusion–reaction model

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left( D \frac{\partial C}{\partial x} \right) + \frac{\partial}{\partial y} \left( D \frac{\partial C}{\partial y} \right) - k_{\text{ann}} C + S$$

## Thermally activated coefficients

$$D(T) = D_0 \exp \left( -\frac{E_D}{k_B T} \right),$$

$$k_{\text{ann}}(T) = k_0 \exp \left( -\frac{E_{\text{ann}}}{k_B T} \right).$$

- ▶  $D$  controls spatial migration;  $k_{\text{ann}}$  controls local defect removal or transformation.
- ▶ Higher  $T$  generally increases both migration and annealing by reducing the Boltzmann penalty for crossing activation barriers.
- ▶ Important timescales:

$$\tau_D \sim \frac{L^2}{D}, \quad \tau_{\text{ann}} \sim \frac{1}{k_{\text{ann}}}.$$

- ▶ If  $\tau_D \ll \tau_{\text{ann}}$ , defects redistribute before disappearing; if  $\tau_D \gg \tau_{\text{ann}}$ , they anneal locally before spreading.
- ▶ Default closed-domain boundary:

$$\hat{\mathbf{n}} \cdot \mathbf{J}_C = 0, \quad \hat{\mathbf{n}} \cdot D \nabla C = 0.$$

# Finite-element form, update equation, and verification checks

Weak form for the 2-D defect-evolution equation

$$\int_{\Omega} w \frac{\partial C}{\partial t} d\Omega + \int_{\Omega} D \nabla w \cdot \nabla C d\Omega + \int_{\Omega} w k_{\text{ann}} C d\Omega = \int_{\Omega} w S d\Omega$$

$$C_h(x, y, t) = \sum_{j=1}^N C_j(t) N_j(x, y),$$

$$M_{ij} = \int_{\Omega} N_i N_j d\Omega,$$

$$(K_D)_{ij} = \int_{\Omega} D \nabla N_i \cdot \nabla N_j d\Omega,$$

$$(K_R)_{ij} = \int_{\Omega} k_{\text{ann}} N_i N_j d\Omega.$$

Backward-Euler update

$$[M + \Delta t(K_D + K_R)] \mathbf{C}^{m+1} = M \mathbf{C}^m + \Delta t \mathbf{s}^{m+1}$$

- ▶ Main output:  $C(x, y, t)$ , which feeds defect charge in Module 2 and defect-dependent coefficients in later modules.
- ▶ Pure annealing check:  $D = 0 \Rightarrow C(t) = C(0)e^{-k_{\text{ann}} t}$ .
- ▶ Pure diffusion with zero-flux boundaries:  $R = 0 \Rightarrow M(t) = \int_{\Omega} C d\Omega$  should be conserved.
- ▶ Uniform initial state should remain uniform if sources and boundaries preserve uniformity.

## Module 4 purpose: thermal response beyond pure Fourier diffusion

Module 4 computes the 2-D temperature or medium-energy field driven by radiation heating and later coupled electrical heating.

$$\{\dot{q}_{\text{rad}}, Q_J, Q_{\text{rec}}, Q_{\text{def}}, \text{thermal BCs}\} \longrightarrow T(x, y, t) \longrightarrow \{D_i, k_{\text{ann},i}, \mu_n, \mu_p, R\}$$

### Conservation of thermal energy

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{q} = \dot{q}, \quad u = CT \quad \text{for constant volumetric heat capacity } C.$$

- ▶  $u(x, y, t)$  is internal energy density;  $T(x, y, t)$  is the equivalent local temperature.
- ▶  $\dot{q}$  may represent thermalized radiation energy deposition, Joule heating, recombination heat, or defect-reconfiguration heat.
- ▶ Module 4 feeds temperature-dependent coefficients back to defect evolution, carrier transport, and coupled degradation metrics.
- ▶ The motivation for ballistic-diffusive physics is that localized, short-time heating can propagate with finite carrier flight time before reducing to ordinary diffusion.

# Fourier, Cattaneo, and ballistic-diffusive thermal transport

Fourier conduction is the local diffusive limit.

$$\mathbf{q}_F = -k \nabla T,$$
$$C \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q}.$$

Cattaneo adds flux relaxation and finite thermal-wave speed.

$$\tau \frac{\partial \mathbf{q}}{\partial t} + \mathbf{q} = -k \nabla T,$$
$$\tau C \frac{\partial^2 T}{\partial t^2} + C \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q} + \tau \frac{\partial \dot{q}}{\partial t}.$$

- ▶ Fourier is accurate when the mean free path is much smaller than the device length.
- ▶ Cattaneo fixes the infinite-propagation-speed artifact but is still a local closure.
- ▶ Ballistic-diffusive modeling splits heat transport into an unscattered ballistic channel and a scattered medium channel:

$$u = u_b + u_m, \quad \mathbf{q} = \mathbf{q}_b + \mathbf{q}_m.$$

- ▶ The key dimensionless indicator is the Knudsen number  $\text{Kn} = \Lambda/L$ .



## Module 4 target equation and coupling inputs

### Reduced 2-D ballistic-diffusive medium-temperature equation

$$\tau C \frac{\partial^2 T_m}{\partial t^2} + C \frac{\partial T_m}{\partial t} = \nabla \cdot (k \nabla T_m) - \nabla \cdot \mathbf{q}_b + \left( \dot{q}_e + \tau \frac{\partial \dot{q}_e}{\partial t} \right)$$

$$\dot{q}_e = \dot{q}_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}.$$

- ▶  $\mathbf{q}_b$  is not a claim that the whole device is ballistic; it represents a retained finite-flight-time correction.
- ▶ In the diffusive limit,  $\tau \rightarrow 0$  and  $\mathbf{q}_b \rightarrow 0$ , the equation reduces to the ordinary heat equation.
- ▶ Temperature modifies Module 3 defect kinetics:

$$D_i(T) = D_{i0} \exp\left(-\frac{E_{D,i}}{k_B T}\right), \quad k_{\text{ann},i}(T) = k_{i0} \exp\left(-\frac{E_{\text{ann},i}}{k_B T}\right).$$

- ▶ Temperature also modifies carrier mobilities, recombination rates, and later the strength of Joule-heating feedback.

## Module 4 finite-element update and verification logic

**Simplified Fourier-limit FEM form used as a baseline thermal path**

$$\int_{\Omega} w C \frac{\partial T}{\partial t} d\Omega + \int_{\Omega} k \nabla w \cdot \nabla T d\Omega = \int_{\Omega} w \dot{q} d\Omega + \int_{\Gamma_N} w q_N d\Gamma.$$

$$T_h = \sum_j \Theta_j N_j,$$

$$M_{ij}^T = \int_{\Omega} C N_i N_j d\Omega,$$

$$K_{ij}^T = \int_{\Omega} k \nabla N_i \cdot \nabla N_j d\Omega,$$

$$\left[ M^T + \Delta t K^T \right] \Theta^{m+1} = M^T \Theta^m + \Delta t \mathbf{f}_T^{m+1}.$$

- ▶ Dirichlet boundaries impose fixed temperature; Neumann boundaries impose heat flux; Robin boundaries impose convective exchange.
- ▶ The full Module 4 target keeps ballistic-memory terms, but the Fourier-limit solve is the necessary first verification baseline.
- ▶ Verification checks: uniform equilibrium preservation, diffusive-limit recovery, localized-pulse spreading, and sensitivity to finite-flight-time corrections.
- ▶ Main output:  $T(x, y, t)$  on the same 2-D domain used by Modules 2, 3, 5, and 6.

## Module 5 purpose: carrier transport in the damaged device

Module 5 evolves electron and hole densities under electric fields, diffusion, generation, recombination, and defect-modified material properties.

$$\{\phi, \mathbf{E}, C_i, T, G, R, \mu_n, \mu_p, D_n, D_p\} \longrightarrow n(x, y, t), p(x, y, t), \mathbf{J}_n, \mathbf{J}_p$$

Particle balance laws

$$\begin{aligned}\frac{\partial n}{\partial t} &= \frac{1}{q} \nabla \cdot \mathbf{J}_n + G - R, \\ \frac{\partial p}{\partial t} &= -\frac{1}{q} \nabla \cdot \mathbf{J}_p + G - R.\end{aligned}$$

- ▶ Electrons and holes have opposite charge, so their current-divergence signs differ in the continuity equations.
- ▶  $G$  creates electron-hole pairs;  $R$  removes them through recombination.
- ▶ The electric field comes from Module 2, while defect and temperature fields from Modules 3–4 modify the transport coefficients.
- ▶ Module 5 produces carrier fields that feed back into the Poisson charge density in Module 6.

# Drift-diffusion currents and 2-D carrier PDEs

## Current-density closures

$$\mathbf{J}_n = q\mu_n n \mathbf{E} + qD_n \nabla n,$$

$$\mathbf{J}_p = q\mu_p p \mathbf{E} - qD_p \nabla p.$$

## Drift-diffusion PDE system

$$\frac{\partial n}{\partial t} = \nabla \cdot (\mu_n n \mathbf{E} + D_n \nabla n) + G - R,$$

$$\frac{\partial p}{\partial t} = -\nabla \cdot (\mu_p p \mathbf{E} - D_p \nabla p) + G - R.$$

- ▶ Drift terms scale with mobility, density, and electric field.
- ▶ Diffusion terms respond to carrier-density gradients.
- ▶ In nondegenerate thermal equilibrium, the Einstein relation gives

$$D_n = \mu_n \frac{k_B T}{q}, \quad D_p = \mu_p \frac{k_B T}{q}.$$

- ▶ Steady-state current response is obtained by setting  $\partial n / \partial t = \partial p / \partial t = 0$ .

# How defects degrade mobility and enhance recombination

## Mobility degradation through additional momentum scattering

$$\begin{aligned}\mu_n &= \frac{q\tau_{m,n}}{m_n^*}, & \frac{1}{\tau_{m,n}} &= \frac{1}{\tau_{m,n}^{\text{lat}}(T)} + \frac{1}{\tau_{m,n}^{\text{dop}}} + \\ & & & + \sum_j \sigma_{m,n}^{(j)} v_{\text{th},n} C_j, \\ \mu_p &= \frac{q\tau_{m,p}}{m_p^*}, & \frac{1}{\tau_{m,p}} &= \frac{1}{\tau_{m,p}^{\text{lat}}(T)} + \frac{1}{\tau_{m,p}^{\text{dop}}} + \sum_j \sigma_{m,p}^{(j)} v_{\text{th},p} C_j.\end{aligned}$$

## Trap-assisted Shockley–Read–Hall recombination

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad \frac{1}{\tau_n} = \sum_j \sigma_{n,j} v_{\text{th}} C_j, \quad \frac{1}{\tau_p} = \sum_j \sigma_{p,j} v_{\text{th}} C_j.$$

- ▶ Increasing defect concentration generally increases scattering and lowers mobility.
- ▶ Defects also act as trap populations, shortening carrier lifetime and increasing nonradiative recombination.
- ▶ These closures are the physical path from radiation damage to degraded current response.

# Module 5 boundaries, FEM form, and outputs

## Representative boundary conditions

Ohmic contacts:  $n = n_c, \quad p = p_c,$

Insulating boundaries:  $\mathbf{J}_n \cdot \hat{\mathbf{n}} = 0, \quad \mathbf{J}_p \cdot \hat{\mathbf{n}} = 0,$

Surface recombination:  $\frac{1}{q} \mathbf{J}_n \cdot \hat{\mathbf{n}} = S_n(n - n_{\text{eq}}), \quad -\frac{1}{q} \mathbf{J}_p \cdot \hat{\mathbf{n}} = S_p(p - p_{\text{eq}}).$

## Electron weak-form structure

$$\int_{\Omega} w \frac{\partial n}{\partial t} d\Omega + \int_{\Omega} \nabla w \cdot (\mu_n n \mathbf{E} + D_n \nabla n) d\Omega = \int_{\Omega} w (G - R) d\Omega + \text{boundary terms}.$$

- ▶ The hole weak form is analogous but carries the opposite drift-diffusion sign convention.
- ▶ Module 5 is nonlinear because current terms contain products such as  $\mu_n n \mathbf{E}$  and recombination depends on  $n$  and  $p$ .
- ▶ Main outputs:  $n, p, \mathbf{J}_n, \mathbf{J}_p$ , recombination map, generation map, and current-voltage or degradation metrics.

## Module 6 purpose: self-consistent multiphysics integration

Module 6 couples the fields from Modules 2–5 on the same 2-D device cross section.

$$\{C_j\} \leftrightarrow \rho \leftrightarrow \phi, \mathbf{E} \leftrightarrow n, p, \mathbf{J}_n, \mathbf{J}_p \leftrightarrow T \leftrightarrow \{C_j\}$$

- ▶ Module 2 supplies electrostatic potential and electric field from total charge density.
- ▶ Module 3 supplies defect concentration maps and defect-dependent charge, traps, and kinetic states.
- ▶ Module 4 supplies temperature and heat-transport response.
- ▶ Module 5 supplies carrier densities, currents, generation, recombination, and current-related degradation outputs.
- ▶ The goal is not four separate simulations, but a mutually consistent set of physical maps at each time level.

# Coupled governing equations in Module 6

## A compact coupled PDE system

$$\begin{aligned}\frac{\partial C_j}{\partial t} &= \nabla \cdot (D_j(T) \nabla C_j) + R_j(\{C\}, n, p, T) + S_j, \\ \nabla \cdot (\epsilon \nabla \phi) &= -q \left[ p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \\ C_T \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}}, \\ \frac{\partial n}{\partial t} &= \nabla \cdot (\mu_n n \mathbf{E} + D_n \nabla n) + G - R, \\ \frac{\partial p}{\partial t} &= -\nabla \cdot (\mu_p p \mathbf{E} - D_p \nabla p) + G - R, \\ \mathbf{E} &= -\nabla \phi.\end{aligned}$$

- ▶ The equations are coupled through coefficients, sources, charge density, electric field, and heat generation.
- ▶ The thermal equation shown here is the compact diffusive version; Module 4 may replace it with the reduced ballistic-diffusive form.



# Main physical feedback loops and staged coupling levels

## Key feedback mechanisms

$$\begin{aligned}\{C_j, z_j\} &\rightarrow \rho_{\text{def}} \rightarrow \phi, \mathbf{E}, \\ \mathbf{E} &\rightarrow \mathbf{J}_n, \mathbf{J}_p \rightarrow n, p, \\ C_j &\rightarrow \mu_n, \mu_p, \tau_n, \tau_p \rightarrow \mathbf{J}_n, \mathbf{J}_p, R, \\ T &\rightarrow D_j, k_{\text{ann},j}, \mu_n, \mu_p, R, \\ \mathbf{J} \cdot \mathbf{E} &\rightarrow Q_J \rightarrow T.\end{aligned}$$

## Recommended physical-coupling ladder

1. Level 0: one-way field transfer among modules.
2. Level 1: electrostatic-carrier feedback,  $(n, p) \leftrightarrow \rho \leftrightarrow \phi, \mathbf{E}$ .
3. Level 2: defect-electrostatic-carrier coupling,  $C_j \rightarrow \rho_{\text{def}}, N_t \rightarrow R_{\text{SRH}}$ .
4. Level 3: thermal feedback through  $T$ -dependent coefficients and current-induced heat.
5. Level 4: full self-consistent iteration among defects, electrostatics, carriers, and temperature.

## Module 6 consistency checks and device-response outputs

### Charge consistency

$$\rho_{\text{check}} = q \left[ p - n + N_D^+ - N_A^- + \sum_j z_j C_j \right], \quad \epsilon_\rho = \frac{\|\rho_{\text{Poisson}} - \rho_{\text{check}}\|_2}{\|\rho_{\text{check}}\|_2 + \rho_{\text{floor}}}.$$

### Energy-source bookkeeping

$$Q_{\text{tot}} = Q_{\text{rad}} + Q_J + Q_{\text{rec}} + Q_{\text{def}} + Q_{\text{ext}}, \quad P_a(t) = \int_{\Omega} Q_a(x, y, t) d\Omega.$$

### Carrier conservation in an insulating domain

$$\frac{d}{dt} \int_{\Omega} n d\Omega = \int_{\Omega} (G - R) d\Omega, \quad \frac{d}{dt} \int_{\Omega} p d\Omega = \int_{\Omega} (G - R) d\Omega.$$

- ▶ These diagnostics catch stale-field coupling, unit mistakes, and sign errors that can look numerically stable but be physically wrong.
- ▶ Device-level outputs may include current, leakage, recombination power, temperature rise, defect inventory, and degradation scores.

## Module 2 PINN: surrogate mapping for Poisson electrostatics

**Goal:** learn the Module 2 input-output map while penalizing violations of electrostatics.

$$\mathcal{G}_\theta : (\mathbf{x}, \boldsymbol{\mu}) \longmapsto \phi_\theta(\mathbf{x}; \boldsymbol{\mu}), \quad \mathbf{E}_\theta = -\nabla_{\mathbf{x}} \phi_\theta$$

- ▶  $\mathbf{x} = (x, y)$  is the spatial coordinate;  $\boldsymbol{\mu}$  contains problem parameters such as bias, defect amplitude, material constants, geometry, and source descriptors.
- ▶ The finite-element solver remains the reference model; the PINN is a fast differentiable surrogate for repeated Module 2 solves.
- ▶ A purely supervised neural network learns from pairs

$$(\rho, \text{BCs}, \boldsymbol{\mu}) \mapsto \phi_{\text{FEM}}.$$

- ▶ A physics-informed neural network also sees the governing equation and boundary conditions through its loss function.

**Central distinction:** the network is not a new electrostatic law; it is an approximate representation constrained to behave like the Poisson model.

## Module 2 PINN: Poisson residual and boundary losses

Strong-form residual used at interior collocation points

$$\mathcal{R}_P(\mathbf{x}; \theta) = \nabla \cdot (\epsilon \nabla \phi_\theta(\mathbf{x})) + \rho(\mathbf{x})$$

For uniform silicon permittivity,

$$\mathcal{R}_P(\mathbf{x}; \theta) = \epsilon_{\text{Si}} \left( \frac{\partial^2 \phi_\theta}{\partial x^2} + \frac{\partial^2 \phi_\theta}{\partial y^2} \right) + \rho(\mathbf{x}).$$

Representative loss terms

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_r} \sum_{k=1}^{N_r} |\mathcal{R}_P(\mathbf{x}_k; \theta)|^2,$$

$$\mathcal{L}_D = \frac{1}{N_D} \sum_{k=1}^{N_D} |\phi_\theta(\mathbf{x}_k) - \phi_D(\mathbf{x}_k)|^2,$$

$$\mathcal{L}_N = \frac{1}{N_N} \sum_{k=1}^{N_N} |-\epsilon \nabla \phi_\theta(\mathbf{x}_k) \cdot \hat{\mathbf{n}} - g_N(\mathbf{x}_k)|^2.$$

## Module 2 PINN: training objective and data construction

### Combined supervised-plus-physics objective

$$\mathcal{L}_{\text{total}} = \lambda_s \mathcal{L}_{\text{data}} + \lambda_r \mathcal{L}_{\text{PDE}} + \lambda_D \mathcal{L}_D + \lambda_N \mathcal{L}_N + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}$$

- Supervised samples come from trusted FEM cases:

$$\mathcal{D}_s = \{(\mathbf{x}_k, \boldsymbol{\mu}_k, \phi_{\text{FEM},k})\}_{k=1}^{N_s}.$$

- Interior residual points do not need FEM labels; they train the model to satisfy Poisson's equation between labeled samples.
- Boundary points enforce contact voltages, ground reference, insulating conditions, or prescribed normal flux.
- Inputs should be nondimensionalized, for example

$$\hat{x} = x/L, \quad \hat{\phi} = \phi/\phi_0, \quad \hat{\rho} = \rho/\rho_0.$$

- The loss weights must be scaled so that the largest-magnitude term does not dominate training for numerical rather than physical reasons.

## Module 2 PINN: verification, strengths, and limits

**A useful PINN surrogate must be judged against the reference electrostatic solve, not only by low training loss.**

- Compare potential and field errors:

$$e_\phi = \frac{\|\phi_\theta - \phi_{\text{FEM}}\|_2}{\|\phi_{\text{FEM}}\|_2 + \phi_{\text{floor}}}, \quad e_E = \frac{\|\mathbf{E}_\theta - \mathbf{E}_{\text{FEM}}\|_2}{\|\mathbf{E}_{\text{FEM}}\|_2 + E_{\text{floor}}}.$$

- Check physically simple cases: zero charge, uniform space charge, linear potential, localized defect charge, and boundary-condition consistency.
- Strength: once trained, the surrogate may accelerate parameter sweeps and sensitivity scans.
- Limitation: a PINN can still satisfy a weakly weighted residual while violating charge conservation, boundary behavior, or sharp field gradients.
- Practical conclusion: keep FEM as the truth model; use the PINN only after systematic out-of-distribution testing.

## Module 2 CPINN: why a causal training curriculum is needed

Module 2 is elliptic and quasi-static, so causality does not mean electrostatic time evolution.

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho, \quad \mathbf{E} = -\nabla \phi.$$

- ▶ A single Poisson solve has no intrinsic marching direction in time.
- ▶ Causal training becomes meaningful only when Module 2 is embedded in an ordered sequence such as dose, irradiation time, defect amplitude, boundary loading, or coupled-iteration continuation.
- ▶ The ordered coordinate is denoted  $\tau$  or  $\alpha$ :

$$\rho(\mathbf{x}; \alpha_0), \rho(\mathbf{x}; \alpha_1), \dots, ho(\mathbf{x}; \alpha_K), \quad 0 \leq \alpha_0 < \alpha_1 < \dots < \alpha_K \leq 1.$$

- ▶ The CPINN objective is to learn easy or early states before enforcing later, harder, more highly loaded states.

**Interpretation:** Module 2 CPINN is a causality-respecting curriculum for a family of quasi-static Poisson problems.

## Module 2 CPINN: continuation coordinate and staged residuals

A convenient continuation path from easy to full electrostatic loading

$$\rho(\mathbf{x}; \alpha) = (1 - \alpha)\rho_{\text{easy}}(\mathbf{x}) + \alpha\rho_{\text{full}}(\mathbf{x}), \quad 0 \leq \alpha \leq 1$$

Residual at each continuation slice

$$\mathcal{R}_P(\mathbf{x}, \alpha; \theta) = \nabla \cdot (\epsilon \nabla \phi_\theta(\mathbf{x}, \alpha)) + \rho(\mathbf{x}; \alpha).$$

Slice loss

$$\mathcal{L}_k = \frac{1}{N_k} \sum_{m=1}^{N_k} |\mathcal{R}_P(\mathbf{x}_m, \alpha_k; \theta)|^2 + \mathcal{L}_{D,k} + \mathcal{L}_{N,k} + \mathcal{L}_{\text{data},k}.$$

- ▶  $\alpha = 0$  may represent zero defect charge, weak source amplitude, simple boundary loading, or an earlier dose state.
- ▶  $\alpha = 1$  is the full target electrostatic problem.
- ▶ Staged training prevents the network from fitting late/high-damage states before it can satisfy simpler earlier states.



## Module 2 CPINN: causal weighting of ordered losses

One causal-PINN strategy weights later residuals by earlier residual success.

$$\mathcal{L}_{\text{causal}} = \sum_{k=0}^K w_k \mathcal{L}_k, \quad w_k = \exp \left[ -\eta \sum_{j=0}^{k-1} \mathcal{L}_j \right]$$

- ▶ If early slices have large residuals, then  $w_k$  suppresses the influence of later slices.
- ▶ Once early slices are learned, later slices become active because their causal weights increase.
- ▶ This is not a physical modification of Poisson's equation; it is a training schedule imposed on the learning objective.
- ▶ The full loss may be written as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{causal}} + \lambda_D \mathcal{L}_D + \lambda_N \mathcal{L}_N + \lambda_s \mathcal{L}_{\text{data}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}.$$

- ▶ The parameter  $\eta$  controls how aggressively the curriculum enforces earlier states first.

## Module 2 CPINN: project use and verification criteria

**The CPINN should be accepted only if it improves ordered-state learning without degrading electrostatic fidelity.**

- ▶ Compare against the ordinary PINN at every ordered slice:

$$e_{\phi}(\alpha_k) = \frac{\|\phi_{\theta}(\cdot, \alpha_k) - \phi_{\text{FEM}}(\cdot, \alpha_k)\|_2}{\|\phi_{\text{FEM}}(\cdot, \alpha_k)\|_2 + \phi_{\text{floor}}}.$$

- ▶ Check whether late/high-damage states improve because early/low-damage states were learned first.
- ▶ Verify that boundary errors do not grow at large  $\alpha$ .
- ▶ Verify that the Poisson residual decreases across the whole continuation path, not only at the final state.
- ▶ Useful ordered coordinates for this project include dose, defect source amplitude, applied bias, and coupled-iteration loading.

# Module 11 GNN: target physical dependency graph

Module 11 learns directed relations among physical variables, not mesh adjacency or numerical solver order.

$$\mathcal{G}_{\text{phys}} = (\mathcal{V}_{\text{phys}}, \mathcal{E}_{\text{phys}}), \quad X_i \rightarrow X_j \iff X_i \text{ directly enters the law for } X_j.$$

- Nodes include fields, coefficients, sources, and state variables from Modules 2–5:

$$C_j, \rho, \phi, \mathbf{E}, T, n, p, \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p, R, \mathbf{J}_n, \mathbf{J}_p, Q_J, Q_{\text{rad}}.$$

- Edges are directed physical dependencies: algebraic, constitutive, source, reaction, transport, or delayed dynamical effects.
- The learned graph is allowed to contain cycles because real coupled physics contains feedback.
- The goal is edge discovery, mechanism typing, strength estimation, lag classification, and uncertainty quantification.

# Module 11 GNN: typed edges and physical meaning

Each candidate edge carries more information than present/absent.

$$X_i \rightarrow X_j \mapsto (A_{ij}, z_{ij}, s_{ij}, \ell_{ij}, \sigma_{ij})$$

- ▶  $A_{ij}$ : learned edge strength or probability.
- ▶  $z_{ij}$ : mechanism type, such as algebraic, constitutive, source, transport, reaction, derivative, or delayed dynamical edge.
- ▶  $s_{ij}$ : sign or monotonic trend when physically meaningful.
- ▶  $\ell_{ij}$ : lag class, distinguishing instantaneous relations from delayed responses.
- ▶  $\sigma_{ij}$ : uncertainty or confidence in the inferred edge.

Examples of target relations

$$\begin{aligned} C_j &\rightarrow \rho_{\text{def}}, \\ C_j &\rightarrow \mu_n, \mu_p, \\ \mathbf{E}, n, p &\rightarrow \mathbf{J}_n, \mathbf{J}_p, \end{aligned}$$

$$\begin{aligned} \rho &\rightarrow \phi, \\ C_j &\rightarrow \tau_n, \tau_p, \\ \mathbf{J} \cdot \mathbf{E} &\rightarrow Q_J, \end{aligned}$$

$$\begin{aligned} \phi &\rightarrow \mathbf{E}, \\ T &\rightarrow D_j, k_{\text{ann},j}, \\ Q_J &\rightarrow T. \end{aligned}$$

# Module 11 GNN: message passing with latent adjacency

A variable-level GNN updates each physical node using messages from candidate parent variables.

$$\begin{aligned}\mathbf{m}_{ij}^{(\ell)} &= A_{ij} M_{z_{ij}}\left(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}, \mathbf{g}\right), \\ \bar{\mathbf{m}}_j^{(\ell)} &= \sum_{i \neq j} \mathbf{m}_{ij}^{(\ell)}, \\ \mathbf{h}_j^{(\ell+1)} &= U\left(\mathbf{h}_j^{(\ell)}, \bar{\mathbf{m}}_j^{(\ell)}, \mathbf{g}\right).\end{aligned}$$

- ▶  $\mathbf{h}_j$  is the learned representation of physical variable  $X_j$ .
- ▶  $\mathbf{e}_{ij}$  stores candidate-edge metadata such as allowed mechanism class, dimensional compatibility, and intervention labels.
- ▶  $\mathbf{g}$  stores global context such as radiation condition, bias, material identity, temperature scale, and time index.
- ▶ The adjacency variable  $A_{ij}$  can be supervised when the synthetic generator knows the true graph, or treated as a latent variable when only trajectories are available.
- ▶ Sparsity is essential: without a penalty or prior, the model may explain everything using too many weak edges.

# Module 11 GNN: synthetic trajectories and interventions

Synthetic data must vary both parameter values and graph structure.

$$\mathcal{D}^{(r)} = \left\{ \boldsymbol{\theta}^{(r)}, A^{(r)}, Z^{(r)}, L^{(r)}, X^{(r)}(t), I^{(r)} \right\}.$$

- ▶  $\boldsymbol{\theta}^{(r)}$  contains material, source, transport, trapping, thermal, and boundary parameters.
- ▶  $A^{(r)}$  is the ground-truth physical adjacency used to generate the case.
- ▶ Mechanism switches create graph-family variation:

$$s_{ij} \in \{0, 1\}, \quad \gamma_{ij} \sim \text{log-uniform or bounded random strength}.$$

- ▶ Matched interventions isolate edges by changing one mechanism while keeping the same random seed:

$$z_j = 0, \quad \alpha_{\mu, C} = 0, \quad \alpha_{\tau, C} = 0, \quad \gamma_J = 0, \quad T(\mathbf{x}, t) = T_0.$$

- ▶ Training, validation, and test splits should be by complete physical case, not by time samples from the same trajectory.

## Module 11 GNN: from learned graph to physical coupling sequence

The learned graph may contain feedback, so the output is a partial physical ordering, not a single serial algorithm.

$$\mathcal{G}_{\text{phys}} \longrightarrow \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K\} \longrightarrow \mathcal{G}_{\text{cond}}.$$

- ▶ Strongly connected components  $\mathcal{C}_k$  collect mutually dependent feedback groups.
- ▶ Collapsing each feedback group into one supernode gives the condensation graph  $\mathcal{G}_{\text{cond}}$ , which is acyclic.
- ▶ A topological ordering of  $\mathcal{G}_{\text{cond}}$  gives a physically admissible coupling sequence among influence groups:

$$\mathcal{C}_{\pi(1)} \prec \mathcal{C}_{\pi(2)} \prec \dots \prec \mathcal{C}_{\pi(K)}.$$

- ▶ Instantaneous algebraic constraints, such as  $\rho \rightarrow \phi \rightarrow \mathbf{E}$ , should be separated from delayed dynamical effects, such as  $T \rightarrow D_j \rightarrow C_j(t + \Delta t)$ .
- ▶ Success means recovering known synthetic edges, rejecting ablated edges, quantifying edge uncertainty, and identifying regime-dependent feedback groups.